Anchored attention converges in polynomial time: a convergence theorem for a single layer/head of a reasoning model

1 Notation and setup

We collect the notation, the formal setup, and the standard arithmetic conventions used throughout the rest of the paper.

Notation. We work in $\mathbb{R}^d$ with the Euclidean norm $\|\cdot\|$ and the inner product $\langle \cdot, \cdot \rangle$. Throughout, $c, C, C_1, C_2, \ldots$ denote universal positive constants whose values may change from line to line and depend only on quantities specified in each lemma's hypothesis. Constants that depend on additional problem parameters carry those parameters as subscripts (e.g. $C_{B,M}$).

We fix a finite set F of *questions* (the "question field"), a single transformer layer index i and a single attention-head index h within that layer; these are fixed throughout the paper. The argument extends to any other (i, h) pair by union bound at logarithmic cost in the total number of (layer, head) pairs (see Remark 10.7 in Section 10).

The reasoning trajectory. For a question $Q \in F$, the reasoning model generates a sequence of tokens between the markers $\langle$think$\rangle$ and $\langle$/think$\rangle$. We index the generated tokens by $j = 1, 2, \ldots$ in emission order. We write $\mathcal{F}_{j-1}$ for the σ-algebra generated by the model's stochastic sampling decisions up to (but not including) step j; the sampling at step j is then $\mathcal{F}_{j-1}$-conditional.

For each $j \geq 1$, let $k_j \in \mathbb{R}^{d_k}$ and $V_j \in \mathbb{R}^d$ denote the *key* and *value* vectors produced by layer i, head h at the j-th reasoning position, and let $q \in \mathbb{R}^{d_k}$ denote the query vector at the $\langle$/think$\rangle$ position (the position at which the model commits to an answer). The key, value, and query vectors depend on the model parameters (fixed) and on Q and the reasoning trajectory generated so far.

Definition 1.1 (Cumulative softmax attention)**.** The single-head softmax attention output of layer i, head h, at the $\langle$/think$\rangle$ position, restricted to the first j reasoning keys, is

$$x_j \;:=\; \sum_{k=1}^{j} w_{j,k} V_k, \qquad w_{j,k} \;:=\; \frac{e^{\langle q, k_k \rangle}}{s_j}, \qquad s_j \;:=\; \sum_{i=1}^{j} e^{\langle q, k_i \rangle}, \tag{1}$$

where $\langle q, k_k \rangle$ denotes the rescaled inner product $q^\top k_k / \sqrt{d_k}$ absorbing the standard $\sqrt{d_k}$ normalisation factor of [6].

Remark 1.2. With $x_0 := 0$, the recurrence form $x_j = (s_{j-1}/s_j)\, x_{j-1} + (e^{\langle q, k_j \rangle}/s_j)\, V_j$ is equivalent to Definition 1.1 (see Lemma 3.1 for the formal identity). This is the shape that motivates the "one-step update" framing of the x_j-dynamics.

Vocabulary and anchors. Let $\mathcal{V}$ denote the model's tokenizer vocabulary, a fixed finite set of tokens. The reasoning trajectory is a sequence of tokens $a_1, a_2, \ldots \in \mathcal{V}$ (with a_j the token emitted at step j). We will tag certain $\mathcal{F}_{j-1}$-conditional events on a_j as "the trajectory emitted an anchor token at step j"; the anchor set itself is question-conditional and is fixed in Assumption 2.1 below.

Decoding. The model's decoding map, applied to a latent vector $x \in \mathbb{R}^d$, returns a final answer token (or token sequence) $\mathrm{decode}(x) \in \mathcal{V}^*$. We treat $\mathrm{decode}(\cdot)$ as a black-box deterministic function of x, and write $\mathrm{Correct}(Q) \subseteq \mathcal{V}^*$ for the (non-empty) set of answer strings that are correct for question Q.

2 Assumptions

The convergence theorem rests on five assumptions: a value-accuracy condition on the question-conditional anchor set (Assumption 2.1), a positive-emission probability for anchors (Assumption 2.3), a score margin between anchor and non-anchor keys (Assumption 2.5), uniform boundedness of value vectors (Assumption 2.7), and existence of a decoding margin around the question-conditional target (Assumption 2.9). Each assumption is accompanied by one remark commenting on its mildness, testability, and how it could be weakened.

Assumption 2.1 (Anchor-set value accuracy)**.** There exist constants $\varepsilon_{\mathrm{anc}} \geq 0$ and $\gamma_{\min} > 0$ such that, for every question $Q \in F$, there is a non-empty question-conditional anchor set $\mathcal{A}(Q) \subseteq \mathcal{V}$ (a subset of the tokenizer vocabulary, depending on Q but *not* on the reasoning trajectory) and a vector $V^*(Q) \in \mathbb{R}^d$ with

$$\max_{a \in \mathcal{A}(Q)} \|V(a) - V^*(Q)\| \;\leq\; \varepsilon_{\mathrm{anc}}. \tag{2}$$

Here $V(a) \in \mathbb{R}^d$ is the layer-i, head-h value-projection vector of the token a when a is decoded as the current reasoning token under Q.[1] We further assume $\gamma(Q) > 2\varepsilon_{\mathrm{anc}}$ for every $Q \in F$, where $\gamma(Q)$ is the decoding margin of Assumption 2.9; we write $\gamma_{\min} := \min_{Q \in F} \gamma(Q) > 0$.

Remark 2.2. Assumption 2.1 is the central sellable assumption. It says: for each question, the model has learned a set of vocabulary tokens whose layer-i/head-h value projections cluster within $\varepsilon_{\mathrm{anc}}$ of a single target $V^*(Q)$. The set $\mathcal{A}(Q)$ is not required to be unique, exhaustive, or constructively specifiable; the proof only needs that some such set exists and that $\varepsilon_{\mathrm{anc}}$ is bounded uniformly across F. Examples of natural anchor sets: synonyms of the correct answer, tokens that appear in the chain-of-thought of a verified solution, or simply $\mathcal{A}(Q) =$ the set of tokens whose embedding decodes to a string in $\mathrm{Correct}(Q)$. The assumption is testable from inference-time observations alone: sample value vectors $V(a)$ at the relevant layer/head for $a \in \mathcal{A}(Q)$ across a held-out set of trajectories, and verify that they cluster.

Assumption 2.3 (Conditional anchor-emission probability)**.** There exists a constant $p_0 \in (0, 1]$ such that, for every $Q \in F$ and every reasoning step $j \geq 1$,

$$\mathbb{P}\big[\, a_j \in \mathcal{A}(Q) \,\big|\, \mathcal{F}_{j-1} \,\big] \;\geq\; p_0, \tag{3}$$

almost surely on the event that Q is the active question. That is, conditional on the history $\mathcal{F}_{j-1}$ of the trajectory, the reasoning policy emits an anchor token at step j with probability at least p_0, uniformly in j and Q.

Remark 2.4. Assumption 2.3 is satisfied if the reasoning policy re-mentions answer-relevant tokens with positive frequency at every step, which is empirically the dominant mode of chain-of-thought

[1] $V(\cdot)$ depends on the trajectory through positional embeddings and prior context. We assume this dependence is mild enough that $\|V(a) - V^*(Q)\| \leq \varepsilon_{\mathrm{anc}}$ holds whenever $a \in \mathcal{A}(Q)$, uniformly across the trajectories the reasoning policy can emit. See Remark 2.2.

reasoning in modern thinking models [8, 4, 2, 5]. The lower bound p_0 can be small (e.g. $p_0 = 0.05$ in practice) and need not be tight; the polynomial-horizon bound Theorem 10.1 depends on $1/p_0$ polynomially.

Assumption 2.5 (Score margin)**.** There exists a constant $\Delta > 0$ such that, for every step $j \geq 1$ and every pair of tokens (a, a') with $a \in \mathcal{A}(Q)$ and $a' \notin \mathcal{A}(Q)$, the softmax pre-activation scores at the $\langle$/think$\rangle$ position satisfy

$$\langle q, k(a)\rangle \;-\; \langle q, k(a')\rangle \;\geq\; \Delta, \tag{4}$$

where $k(a) \in \mathbb{R}^{d_k}$ is the layer-i, head-h key vector that the model produces when a is the token at the current reasoning position. The margin Δ is uniform in j and in $Q \in F$.

Remark 2.6. Assumption 2.5 says the attention mechanism systematically upweights anchor positions over non-anchor positions. This is a "trained-model" condition: it must hold for the fixed weights of layer i, head h. Empirically, attention heads in reasoning models do exhibit strong inductive biases toward answer-relevant tokens (this is the standard observation behind "attention pattern analysis"); the margin Δ can be estimated from the post-softmax weight distribution on held-out trajectories. The dependence of the convergence horizon on Δ is polynomial in $e^{-\Delta}$.

Assumption 2.7 (Bounded value norms)**.** There exists a constant $M > 0$ such that the layer-i, head-h value projection satisfies $\|V(a)\| \leq M$ for every token $a \in \mathcal{V}$. Consequently $\|V_j\| \leq M$ for every reasoning step j.

Remark 2.8. Assumption 2.7 is mild: the model parameters and the token embeddings are bounded; LayerNorm and finite-precision implementations guarantee $\|V(a)\|$ stays in a finite range. The constant M can be computed exactly for a given model by enumerating $\mathcal{V}$. The convergence bound depends on M polynomially.

Assumption 2.9 (Decoding margin (existence))**.** For every $Q \in F$, there exist a vector $V^*(Q) \in \mathbb{R}^d$ (the same $V^*(Q)$ as in Assumption 2.1) and a constant $\gamma(Q) > 0$ such that, for every $x \in \mathbb{R}^d$,

$$\|x - V^*(Q)\| \;\leq\; \gamma(Q) \quad\Longrightarrow\quad \mathrm{decode}(x) \in \mathrm{Correct}(Q). \tag{5}$$

Remark 2.10. Assumption 2.9 is an *existence* statement, not a constructive one. It says only that *some* ball around the question-conditional target $V^*(Q)$ exists whose decoded output lies in the correct-answer set $\mathrm{Correct}(Q)$. The size $\gamma(Q)$ of the ball is question-dependent and may be small in absolute terms — what matters is that it is strictly positive, and (per Assumption 2.1) that it exceeds $2\varepsilon_{\mathrm{anc}}$ uniformly. The assumption does not require knowing $\gamma(Q)$ at inference time or constructing the ball explicitly; it is purely an existence statement about the geometry of the decoding map.

3 The softmax recurrence as a running weighted average

Lemma 3.1 (Softmax recurrence as running average)**.** *Let x_j, $w_{j,k}$, and s_j be defined as in Definition 1.1. With the convention $x_0 := 0$ and $s_0 := 0$, the sequence $(x_j)_{j\geq 0}$ satisfies, for every $j \geq 1$, both the recurrence*

$$x_j \;=\; \frac{s_{j-1}}{s_j}\, x_{j-1} \;+\; \frac{e^{\langle q, k_j\rangle}}{s_j}\, V_j, \tag{6}$$

and the convex-combination identity

$$x_j \;=\; \sum_{k=1}^{j} w_{j,k}\, V_k, \qquad w_{j,k} \;\geq\; 0, \qquad \sum_{k=1}^{j} w_{j,k} \;=\; 1. \tag{7}$$

Proof. The proof is a one-line algebraic identity followed by induction on j.
Step 1: recurrence. By the definition $s_j = \sum_{i=1}^{j} e^{\langle q,k_i\rangle}$, we have $s_j = s_{j-1} + e^{\langle q,k_j\rangle}$ for every $j \geq 1$. Multiplying Eq. (1) by s_j gives

$$s_j\, x_j \;=\; \sum_{k=1}^{j} e^{\langle q,k_k\rangle}\, V_k \;=\; \sum_{k=1}^{j-1} e^{\langle q,k_k\rangle}\, V_k \;+\; e^{\langle q,k_j\rangle}\, V_j \;=\; s_{j-1}\, x_{j-1} \;+\; e^{\langle q,k_j\rangle}\, V_j,$$

where the last equality used $s_{j-1}x_{j-1} = \sum_{k=1}^{j-1} e^{\langle q,k_k\rangle} V_k$ from Eq. (1) at index $j-1$ (and the conventions $x_0 = 0$, $s_0 = 0$ from the lemma statement). Dividing both sides by $s_j > 0$ yields Eq. (6).
Step 2: convex weights. The weights $w_{j,k} = e^{\langle q,k_k\rangle}/s_j$ are non-negative since exponentials are positive and $s_j > 0$. Their row sum is

$$\sum_{k=1}^{j} w_{j,k} \;=\; \frac{1}{s_j} \sum_{k=1}^{j} e^{\langle q,k_k\rangle} \;=\; \frac{s_j}{s_j} \;=\; 1,$$

which establishes Eq. (7) directly from Eq. (1). This completes the proof. □

4 Anchor-set decomposition of the weighted average

Throughout this section we fix an arbitrary $Q \in F$ and write $V^* = V^*(Q)$ for brevity. The following lemma is a triangle-inequality decomposition that splits the error $\|x_j - V^*\|$ into an anchor part and a non-anchor leakage part, parameterized by any subset $\mathcal{A} \subseteq \{1, \ldots, j\}$. We will instantiate $\mathcal{A}$ in subsequent sections with the trajectory-defined anchor index set

$$\mathcal{A}_j^{\mathrm{traj}} \;\coloneqq\; \big\{\, k \in \{1, \ldots, j\} \,:\, a_k \in \mathcal{A}(Q) \,\big\}, \tag{8}$$

i.e. the set of trajectory positions at which the emitted token belongs to the anchor set $\mathcal{A}(Q)$ of Assumption 2.1.

Lemma 4.1 (Anchor decomposition)**.** *Let $j \geq 1$, let $w_{j,k} \geq 0$ with $\sum_{k=1}^{j} w_{j,k} = 1$, and let $V_1, \ldots, V_j, V^* \in \mathbb{R}^d$. For any subset $\mathcal{A} \subseteq \{1, \ldots, j\}$,*

$$\|[\| \Big] \sum_{k=1}^{j} w_{j,k}\, V_k \;-\; V^* \;\leq\; \sum_{k \in \mathcal{A}} w_{j,k}\, \|V_k - V^*\| \;+\; \Big(1 - \sum_{k \in \mathcal{A}} w_{j,k}\Big) \cdot D_j, \tag{9}$$

where $D_j \coloneqq \max_{k \in \{1,\ldots,j\}} \|V_k - V^*\|$.

Proof. The proof is a one-display algebraic identity followed by the triangle inequality and convex-combination bound.

Because the weights $(w_{j,k})_{k=1}^{j}$ form a probability distribution (i.e. $\sum_k w_{j,k} = 1$), we have $V^* = \sum_{k=1}^{j} w_{j,k} V^*$, so

$$\sum_{k=1}^{j} w_{j,k}\, V_k \;-\; V^* \;=\; \sum_{k=1}^{j} w_{j,k}\, (V_k - V^*).$$

Splitting the index set at $\mathcal{A}$ and applying the triangle inequality,

$$\begin{aligned}
\|[\|\Big]\sum_{k=1}^{j} w_{j,k}\, V_k - V^* \;&\le\; \sum_{k\in\mathcal{A}} w_{j,k}\, \|V_k - V^*\| \;+\; \sum_{k\notin\mathcal{A}} w_{j,k}\, \|V_k - V^*\| \\
&\le\; \sum_{k\in\mathcal{A}} w_{j,k}\, \|V_k - V^*\| \;+\; D_j \cdot \sum_{k\notin\mathcal{A}} w_{j,k} \\
&=\; \sum_{k\in\mathcal{A}} w_{j,k}\, \|V_k - V^*\| \;+\; \Big(1 - \sum_{k\in\mathcal{A}} w_{j,k}\Big) \cdot D_j,
\end{aligned}$$

where the first step follows from the triangle inequality applied term-by-term to the convex combination, the second step follows from $\|V_k - V^*\| \le D_j$ for every $k \in \{1, \dots, j\}$ (the defining property of D_j), and the last step follows from $\sum_{k=1}^{j} w_{j,k} = 1$ rearranged as $\sum_{k\notin\mathcal{A}} w_{j,k} = 1 - \sum_{k\in\mathcal{A}} w_{j,k}$. This gives the assertion. $\square$

5 Anchor accuracy bound

Lemma 5.1 (Anchor accuracy)**.** *Fix $Q \in F$ and let $V^* = V^*(Q)$. Under Assumption 2.1, on the event $\{a_k \in \mathcal{A}(Q)\}$ we have $\|V_k - V^*\| \le \varepsilon_{\mathrm{anc}}$. Consequently, for every $j \ge 1$ and every realisation of the trajectory,*

$$\max_{k\in\mathcal{A}_j^{\mathrm{traj}}} \|V_k - V^*\| \;\le\; \varepsilon_{\mathrm{anc}}. \tag{10}$$

Proof. The proof is a one-line restatement of Assumption 2.1. By definition of V_k, when the trajectory emits token $a_k \in \mathcal{A}(Q)$ at step k, the value vector V_k produced by layer i, head h at the corresponding key position coincides with the layer-i, head-h value-projection $V(a_k)$ of that token. Assumption 2.1 (specifically Eq. (2)) asserts that $\|V(a_k) - V^*(Q)\| \le \varepsilon_{\mathrm{anc}}$ for every $a_k \in \mathcal{A}(Q)$, so $\|V_k - V^*\| \le \varepsilon_{\mathrm{anc}}$. Taking the maximum over $k \in \mathcal{A}_j^{\mathrm{traj}}$ (the set of trajectory positions where this event holds, per Eq. (8)) yields Eq. (10), as desired. $\square$

Remark 5.2. The bound in Lemma 5.1 is deterministic given the trajectory: it holds for every realisation, not just with high probability. The randomness in the argument only enters later, through the size of $\mathcal{A}_j^{\mathrm{traj}}$ in Lemma 6.1 and through the softmax-mass lower bound in Lemma 7.1.

6 High-probability lower bound on the anchor count

We now turn from the deterministic algebra of Lemmas 3.1, 4.1 and 5.1 to the first probabilistic step: the number of anchor emissions among the first T reasoning steps is at least $p_0 T/2$ with high probability. The argument is a multiplicative Chernoff (equivalently, Bernstein–Freedman) bound for sums of $\mathcal{F}_{j-1}$-conditional Bernoulli variables; this gives a sharper exponent than the bounded-difference Azuma–Hoeffding bound because the conditional variance of an anchor-emission indicator is at most p_j rather than the deterministic bound 1.

Lemma 6.1 (Anchor count concentration)**.** *Fix $Q \in F$. Under Assumption 2.3, for every $T \ge 1$ and every $\delta_1 \in (0, 1)$, the event*

$$\mathcal{E}_1(T, \delta_1) \;\coloneqq\; \Big\{ \big|\mathcal{A}_T^{\mathrm{traj}}\big| \;\ge\; \tfrac{p_0 T}{2} \Big\}, \tag{11}$$

satisfies $\mathbb{P}[\mathcal{E}_1(T,\delta_1)] \geq 1-\delta_1$, *provided*

$$T \;\geq\; \frac{8}{p_0}\,\log\frac{1}{\delta_1}. \tag{12}$$

Proof. The proof uses the multiplicative-Chernoff / Bernstein–Freedman bound ([3, Theorem 1.6]; see also the technique digest `chernoff-bernoulli.md`) applied to the centred-indicator sequence of the anchor-emission event. The key saving over Azuma–Hoeffding (the bounded-difference bound used in an earlier draft) is that an indicator $X_j \in \{0,1\}$ with $\mathbb{E}[X_j \mid \mathcal{F}_{j-1}] = p_j$ has conditional variance $p_j(1-p_j) \leq p_j$, which is sharper than the deterministic bounded-difference proxy $c_j^2 = 1$ by a factor of $1/p_j \geq 1/p_0$ in the exponent.

Step 1: martingale construction. For each $j \in \{1,\dots,T\}$, define

$$X_j \;\coloneqq\; \mathbf{1}\big\{a_j \in \mathcal{A}(Q)\big\} \qquad \text{and} \qquad p_j \;\coloneqq\; \mathbb{P}[X_j = 1 \mid \mathcal{F}_{j-1}].$$

By Assumption 2.3, $p_j \geq p_0$ almost surely for every j. Let $D_j \coloneqq X_j - p_j$. Then $\mathbb{E}[D_j \mid \mathcal{F}_{j-1}] = 0$ and $|D_j| \leq 1$ (since $X_j \in \{0,1\}$ and $p_j \in [0,1]$); in addition, the conditional variance satisfies $\mathrm{Var}(D_j \mid \mathcal{F}_{j-1}) = p_j(1-p_j) \leq p_j$. So $M_t \coloneqq \sum_{j=1}^{t} D_j$ is an $(\mathcal{F}_t)$-martingale with $M_0 = 0$.

Step 2: multiplicative-Chernoff bound for conditional Bernoullis. Write $\mu \coloneqq \sum_{j=1}^{T} p_j \geq p_0 T$ a.s., where the inequality is from Assumption 2.3. The multiplicative-Chernoff inequality for $\mathcal{F}_{j-1}$-conditional Bernoulli variables — a special case of Freedman's martingale Bernstein bound [3, Theorem 1.6]; the moment-generating-function proof is by exact computation $\mathbb{E}[e^{-\lambda X_j} \mid \mathcal{F}_{j-1}] = 1 - p_j(1-e^{-\lambda}) \leq \exp(-p_j(1-e^{-\lambda}))$ followed by taking products over j and the standard optimisation $\lambda = \log 2$, see `chernoff-bernoulli.md` — gives

$$\mathbb{P}\Big[\sum_{j=1}^{T} X_j \;\leq\; \tfrac{\mu}{2}\Big] \;\leq\; \exp\Big(-\tfrac{\mu}{8}\Big) \;\leq\; \exp\Big(-\tfrac{p_0 T}{8}\Big), \tag{13}$$

where the last inequality uses $\mu \geq p_0 T$.

Step 3: translate to the anchor-count event. Substituting $\sum_{j=1}^{T} X_j = \big|\mathcal{A}_T^{\mathrm{traj}}\big|$ and using $\mu \geq p_0 T$, the event $\{\sum_j X_j \leq \mu/2\}$ contains $\{\sum_j X_j < p_0 T/2\}$. Therefore $\mathbb{P}[|\mathcal{A}_T^{\mathrm{traj}}| < p_0 T/2] \leq \exp(-p_0 T/8)$.

Step 4: invert the rate. For the right-hand side to be at most δ_1, it suffices that $p_0 T/8 \geq \log(1/\delta_1)$, i.e. $T \geq 8\log(1/\delta_1)/p_0$. This is Eq. (12), and this completes the proof. $\square$

Remark 6.2. An earlier draft of this proof used Azuma–Hoeffding (one-sided bounded-difference) and obtained the weaker rate $\exp(-p_0^2 T/8)$ with horizon $T \geq 8\log(1/\delta_1)/p_0^2$. The sharper rate $\exp(-p_0 T/8)$ stated here improves the exponent by a factor of $1/p_0$, which is the canonical Bernstein-vs-Hoeffding sharpening for sums of Bernoullis (cf. [7, Theorem 2.3]). The improvement is empirically substantive: at the empirically realistic $p_0 \in [0.05, 0.2]$ the test-time horizon needed to reach a given failure probability shrinks by a factor of 5–20. The argument goes through with the same martingale construction; only the concentration step changes from a bounded-difference bound to a conditional-variance bound.

7 Anchor-mass lower bound via the score margin

The decomposition Lemma 4.1 reduces the convergence of x_j to V^* to two questions: (i) how close are the value vectors V_k for $k \in \mathcal{A}_j^{\mathrm{traj}}$ to V^* (answered by Lemma 5.1), and (ii) how much softmax mass do those anchor positions absorb (answered here). The conversion from the score-margin assumption Assumption 2.5 to a softmax-mass lower bound is a one-display exponential algebra step.

Lemma 7.1 (Anchor mass lower bound)**.** *Fix $Q \in F$. Under Assumption 2.5, for every $j \geq 1$ and every realisation of the trajectory with non-empty $\mathcal{A}_j^{\mathrm{traj}} \subseteq \{1, \ldots, j\}$,*

$$\sum_{k \in \mathcal{A}_j^{\mathrm{traj}}} w_{j,k} \;\geq\; 1 \;-\; \frac{\left|\{1, \ldots, j\} \setminus \mathcal{A}_j^{\mathrm{traj}}\right|}{\left|\mathcal{A}_j^{\mathrm{traj}}\right|} \cdot e^{-\Delta}, \tag{14}$$

where $w_{j,k}$ is the softmax weight from Definition 1.1 and $\Delta > 0$ is the score-margin constant of Assumption 2.5.

Proof. The proof proceeds in three steps: a pointwise bound on the exponentiated score gap, a sum-of-non-anchors versus sum-of-anchors ratio bound, and a softmax-normalisation conversion.
Step 1: pointwise exponentiated-score bound. Fix any non-anchor position $\ell \in \{1, \ldots, j\} \setminus \mathcal{A}_j^{\mathrm{traj}}$ and any anchor position $k^\star \in \mathcal{A}_j^{\mathrm{traj}}$. By Assumption 2.5, applied to the pair $(a, a') = (a_{k^\star}, a_\ell)$, $\langle q, k_{k^\star} \rangle - \langle q, k_\ell \rangle \geq \Delta$, so

$$e^{\langle q, k_\ell \rangle} \;\leq\; e^{-\Delta}\, e^{\langle q, k_{k^\star} \rangle}.$$

Averaging this bound over $k^\star \in \mathcal{A}_j^{\mathrm{traj}}$ (the set is non-empty by hypothesis), we obtain

$$e^{\langle q, k_\ell \rangle} \;\leq\; \frac{e^{-\Delta}}{\left|\mathcal{A}_j^{\mathrm{traj}}\right|} \sum_{k \in \mathcal{A}_j^{\mathrm{traj}}} e^{\langle q, k_k \rangle}. \tag{15}$$

Step 2: sum-of-non-anchors versus sum-of-anchors. Summing Eq. (15) over all $\ell \in \{1, \ldots, j\} \setminus \mathcal{A}_j^{\mathrm{traj}}$,

$$\begin{aligned}
\sum_{\ell \notin \mathcal{A}_j^{\mathrm{traj}}} e^{\langle q, k_\ell \rangle} \;&\leq\; \left|\{1, \ldots, j\} \setminus \mathcal{A}_j^{\mathrm{traj}}\right| \cdot \frac{e^{-\Delta}}{\left|\mathcal{A}_j^{\mathrm{traj}}\right|} \sum_{k \in \mathcal{A}_j^{\mathrm{traj}}} e^{\langle q, k_k \rangle} \\
&=\; r \cdot \sum_{k \in \mathcal{A}_j^{\mathrm{traj}}} e^{\langle q, k_k \rangle},
\end{aligned}$$

where the first step follows by summing Eq. (15) over the non-anchor index set, and the last step introduces the shorthand $r := |\{1, \ldots, j\} \setminus \mathcal{A}_j^{\mathrm{traj}}| / |\mathcal{A}_j^{\mathrm{traj}}| \cdot e^{-\Delta}$.
Step 3: convert to softmax mass. Write $A := \sum_{k \in \mathcal{A}_j^{\mathrm{traj}}} e^{\langle q, k_k \rangle}$ and $B := \sum_{\ell \notin \mathcal{A}_j^{\mathrm{traj}}} e^{\langle q, k_\ell \rangle}$. Step 2 reads $B \leq r \cdot A$ and $s_j = A + B$. Therefore

$$\begin{aligned}
\sum_{k \in \mathcal{A}_j^{\mathrm{traj}}} w_{j,k} \;&=\; \frac{A}{A + B} \\
&=\; \frac{1}{1 + B/A} \\
&\geq\; \frac{1}{1 + r} \\
&\geq\; 1 - r,
\end{aligned}$$

where the first step follows from $\sum_{k\in\mathcal{A}_j^{\mathrm{traj}}} w_{j,k} = A/s_j = A/(A+B)$ (by Definition 1.1), the second step is algebraic division, the third step follows from $B/A \le r$ (Step 2) and monotonicity of $x \mapsto 1/(1+x)$, and the last step follows from the elementary inequality $1/(1+r) \ge 1-r$ valid for all $r \ge 0$ (since $(1+r)(1-r) = 1-r^2 \le 1$). Substituting back the definition of r yields Eq. (14), as desired. □

Remark 7.2. The bound Eq. (14) is informative when the anchor-count fraction $|\mathcal{A}_j^{\mathrm{traj}}|/j$ is not vanishing. Combined with the high-probability lower bound $|\mathcal{A}_T^{\mathrm{traj}}| \ge p_0T/2$ from Lemma 6.1, we obtain $|\{1,\dots,T\} \setminus \mathcal{A}_T^{\mathrm{traj}}| \le T$ and $|\mathcal{A}_T^{\mathrm{traj}}| \ge p_0T/2$, so on the event $\mathcal{E}_1(T,\delta_1)$,

$$\sum_{k\in\mathcal{A}_T^{\mathrm{traj}}} w_{T,k} \;\ge\; 1 - \frac{T}{p_0T/2}\cdot e^{-\Delta} \;=\; 1 - \frac{2}{p_0}e^{-\Delta}.$$

For this lower bound to exceed any pre-specified $1-\eta$ with $\eta \in (0,1)$, it suffices that $\Delta \ge \log\bigl(2/(p_0\eta)\bigr)$ — a condition on the trained model's attention biases, not on T.

8 The polynomial horizon

We now combine the lemmas of Sections 4 to 7 into a single high-probability bound on $\|x_T - V^*(Q)\|$. The result isolates the score-margin condition from the time-horizon: the deviation splits into a T-independent leakage term controlled by Δ and a high-probability event controlled by T.

Lemma 8.1 (Polynomial horizon to reach the decoding margin)**.** *Fix $Q \in F$ and a confidence parameter $\delta \in (0,1)$. Suppose Assumptions 2.1, 2.3, 2.5, 2.7 and 2.9 hold and the score-margin constant Δ of Assumption 2.5 satisfies*

$$\Delta \;\ge\; \log\frac{4(M+\|V^*(Q)\|)}{p_0\,\gamma(Q)}. \tag{16}$$

Then for every horizon

$$T \;\ge\; T(Q,\delta) \;:=\; \frac{8}{p_0}\log\frac{2}{\delta}, \tag{17}$$

we have

$$\mathbb{P}\Big[\;\|x_T - V^*(Q)\| \;\le\; \tfrac{\gamma(Q)}{2} \;+\; \varepsilon_{\mathrm{anc}}\Big] \;\ge\; 1 - \tfrac{\delta}{2}. \tag{18}$$

Proof. We combine Lemmas 4.1, 5.1, 6.1 and 7.1. The randomness enters only through Lemma 6.1 (anchor-count concentration); conditional on the event $\mathcal{E}_1 := \{|\mathcal{A}_T^{\mathrm{traj}}| \ge p_0T/2\}$, the remaining algebra is deterministic.

Step 1: anchor-count event. By Lemma 6.1 applied at confidence $\delta_1 = \delta/2$ with hypothesis $T \ge 8\log(2/\delta)/p_0$ (which is Eq. (17)), $\mathbb{P}[\mathcal{E}_1] \ge 1-\delta/2$.

Step 2: combine anchor decomposition with the accuracy and mass bounds. By Lemma 4.1 with the choice $\mathcal{A} = \mathcal{A}_T^{\mathrm{traj}}$,

$$\|x_T - V^*\| \;\le\; \sum_{k\in\mathcal{A}_T^{\mathrm{traj}}} w_{T,k}\;\|V_k - V^*\| \;+\; \Bigl(1 - \sum_{k\in\mathcal{A}_T^{\mathrm{traj}}} w_{T,k}\Bigr)\cdot D_T. \tag{19}$$

By Lemma 5.1, the first summand is bounded by

$$\sum_{k\in\mathcal{A}_T^{\mathrm{traj}}} w_{T,k}\ \|V_k - V^*\| \ \le\ \varepsilon_{\mathrm{anc}}\cdot \sum_{k\in\mathcal{A}_T^{\mathrm{traj}}} w_{T,k}\ \le\ \varepsilon_{\mathrm{anc}}.$$

By the triangle inequality and Assumption 2.7, $D_T = \max_k \|V_k - V^*\| \le \max_k \|V_k\| + \|V^*\| \le M + \|V^*\|$.

By Lemma 7.1 applied on $\mathcal{E}_1$,

$$1 - \sum_{k\in\mathcal{A}_T^{\mathrm{traj}}} w_{T,k}\ \le\ \frac{|\{1,\dots,T\}\setminus\mathcal{A}_T^{\mathrm{traj}}|}{|\mathcal{A}_T^{\mathrm{traj}}|}\cdot e^{-\Delta}\ \le\ \frac{T}{p_0T/2}\cdot e^{-\Delta}\ =\ \frac{2}{p_0}\,e^{-\Delta},$$

where the last inequality uses $|\{1,\dots,T\}\setminus\mathcal{A}_T^{\mathrm{traj}}| \le T$ and $|\mathcal{A}_T^{\mathrm{traj}}| \ge p_0T/2$ (the latter is $\mathcal{E}_1$).

Combining the three pieces, on $\mathcal{E}_1$,

$$\|x_T - V^*\|\ \le\ \varepsilon_{\mathrm{anc}}\ +\ \frac{2}{p_0}\,e^{-\Delta}\cdot(M + \|V^*\|). \tag{20}$$

Step 3: apply the score-margin hypothesis. The hypothesis Eq. (16) is equivalent to $\frac{2}{p_0}e^{-\Delta}(M+\|V^*\|) \le \gamma(Q)/2$. Indeed, $e^{-\Delta} \le p_0\gamma(Q)/(4(M+\|V^*\|))$ rearranges to $\Delta \ge \log(4(M+\|V^*\|)/(p_0\gamma(Q)))$; multiplying by $\frac{2}{p_0}(M+\|V^*\|)$ gives $\frac{2}{p_0}e^{-\Delta}(M+\|V^*\|) \le \gamma(Q)/2$. Substituting into Eq. (20) yields, on $\mathcal{E}_1$,

$$\|x_T - V^*\|\ \le\ \varepsilon_{\mathrm{anc}} + \tfrac{\gamma(Q)}{2}.$$

Step 4: probability budget. Combining Step 1 ($\mathbb{P}[\mathcal{E}_1] \ge 1-\delta/2$) with the deterministic bound on $\mathcal{E}_1$ from Step 3, $\mathbb{P}[\|x_T - V^*\| \le \gamma(Q)/2 + \varepsilon_{\mathrm{anc}}] \ge \mathbb{P}[\mathcal{E}_1] \ge 1-\delta/2$, which is Eq. (18). This completes the proof. □

Remark 8.2. The lemma cleanly separates the two roles of the assumptions: the score-margin condition Eq. (16) is a T-independent property of the trained model, while the polynomial horizon Eq. (17) controls the concentration failure budget. If the trained model's Δ does not satisfy Eq. (16), no horizon T can drive the deviation below $\gamma(Q)/2 + \varepsilon_{\mathrm{anc}}$ in this proof — the leakage term $(2/p_0)e^{-\Delta}(M+\|V^*\|)$ does not shrink with T. The score-margin condition is the load-bearing "trained-model" requirement; the horizon T only suppresses the multiplicative-Chernoff concentration tail.

9 From attention output to correct answer

Lemma 9.1 (Decoding correctness)**.** *Fix $Q \in F$ and let $V^* = V^*(Q)$, $\gamma = \gamma(Q)$ be as in Assumptions 2.1 and 2.9. Under Assumptions 2.1 and 2.9, for every $x \in \mathbb{R}^d$ with*

$$\|x - V^*\|\ \le\ \tfrac{\gamma}{2} + \varepsilon_{\mathrm{anc}}, \tag{21}$$

we have $\|x - V^*\| \le \gamma$ and consequently decode$(x) \in$ Correct(Q).

Proof. The proof is a direct chain of two arithmetic steps.

By the hypothesis $\gamma > 2\varepsilon_{\mathrm{anc}}$ of Assumption 2.1, $\varepsilon_{\mathrm{anc}} \le \gamma/2$. Substituting into Eq. (21),

$$\|x - V^*\|\ \le\ \tfrac{\gamma}{2} + \varepsilon_{\mathrm{anc}}\ \le\ \tfrac{\gamma}{2} + \tfrac{\gamma}{2}\ =\ \gamma.$$

Applying Assumption 2.9 with this x, the implication Eq. (5) yields decode$(x) \in$ Correct(Q), as desired. □

10 The main convergence theorem

We now state and prove the test-time scaling law: the probability that the layer-i, head-h attention output x_T at the $\langle$/think$\rangle$ position fails to land in the decoding-margin ball around $V^*(Q)$ decays exponentially in the reasoning horizon T, with rate $p_0/8$ set by the anchor-emission probability of the policy.

Theorem 10.1 (Test-time scaling for anchored attention). *Suppose Assumptions 2.1, 2.3, 2.5, 2.7 and 2.9 hold, and assume the score margin Δ of Assumption 2.5 satisfies*

$$\Delta \;\geq\; \log \frac{4(M + \max_{Q\in F} \|V^*(Q)\|)}{p_0\,\gamma_{\min}}. \tag{22}$$

(Equivalently, the score margin must be large enough to drive the non-anchor leakage below $\gamma_{\min}/2$.) Then for every $Q \in F$ and every reasoning horizon $T \geq 1$,

$$\mathbb{P}\Big[\; \|x_T - V^*(Q)\| \;\leq\; \tfrac{\gamma(Q)}{2} + \varepsilon_{\mathrm{anc}} \;\; \textit{and} \;\; \mathrm{decode}(x_T) \in \mathrm{Correct}(Q) \Big] \;\geq\; 1 - 2\exp\Big(-\frac{p_0 T}{8}\Big). \tag{23}$$

In particular, the failure probability decays exponentially in the test-time horizon T: $\mathbb{P}[\textit{failure}] \leq 2\exp(-p_0T/8) \to 0$ as $T \to \infty$.

Proof. The proof has three steps: discharge the score-margin hypothesis, identify the single probabilistic event the bound depends on, and read off the rate. We may assume $2\exp(-p_0T/8) < 1$ (else the bound is vacuous), so $\delta := 2\exp(-p_0T/8) \in (0,1)$ is a valid confidence parameter.

Step 1: deterministic-on-event bound via Lemma 8.1. Eq. (22) implies the per-question score-margin condition Eq. (16) for every $Q \in F$, because $\gamma(Q) \geq \gamma_{\min}$ and $\|V^*(Q)\| \leq \max_{Q\in F}\|V^*(Q)\|$ (so the right-hand side of Eq. (22) dominates the right-hand side of Eq. (16)). With $\delta = 2\exp(-p_0T/8)$ as above, $\log(2/\delta) = \log(\exp(p_0T/8)) = p_0T/8$ by construction, so the lemma's horizon $T(Q,\delta) = (8/p_0)\log(2/\delta) = (8/p_0)\cdot(p_0T/8) = T$, and the hypothesis $T \geq T(Q,\delta)$ of Lemma 8.1 holds tautologically. The lemma then gives

$$\mathbb{P}\Big[\; \|x_T - V^*(Q)\| \;\leq\; \tfrac{\gamma(Q)}{2} \;+\; \varepsilon_{\mathrm{anc}} \Big] \;\geq\; 1 - \tfrac{\delta}{2} \;=\; 1 - \exp\Big(-\frac{p_0T}{8}\Big). \tag{24}$$

On the event $\{\|x_T - V^*(Q)\| \leq \gamma(Q)/2 + \varepsilon_{\mathrm{anc}}\}$, Lemma 9.1 gives both $\|x_T - V^*(Q)\| \leq \gamma(Q)$ and $\mathrm{decode}(x_T) \in \mathrm{Correct}(Q)$, so this event is contained in the joint event of Eq. (23).

Step 2: union-bound paragraph. The only probabilistic event used in Step 1 is the anchor-count event $\mathcal{E}_1 := \{|\mathcal{A}_T^{\mathrm{traj}}| \geq p_0T/2\}$ that underlies the proof of Lemma 8.1 via Lemma 6.1. By a union bound over $\mathcal{E}_1$ alone (the only random event with non-zero failure probability), the failure probability of the joint event of Eq. (23) is at most

$$\mathbb{P}[\mathcal{E}_1^c] \;\leq\; \tfrac{\delta}{2} \;=\; \exp\Big(-\frac{p_0T}{8}\Big) \;\leq\; 2\exp\Big(-\frac{p_0T}{8}\Big). \tag{25}$$

Step 3: conclude. Combining Steps 1 and 2, the joint event of Eq. (23) has probability at least $1 - \mathbb{P}[\mathcal{E}_1^c] \geq 1 - 2\exp(-p_0T/8)$, as desired. (The factor 2 is slack absorbed for compatibility with Corollary 10.2; the tighter bound $1 - \exp(-p_0T/8)$ also holds.) When $2\exp(-p_0T/8) \geq 1$, the conclusion holds vacuously since the right-hand side of Eq. (23) is non-positive. $\square$

Corollary 10.2 (Expected-error decay). *Under the assumptions of Theorem 10.1, for every $Q \in F$ and every $T \geq 1$,*

$$\mathbb{E}\big[\, \|x_T - V^*(Q)\| \,\big] \;\leq\; \tfrac{\gamma(Q)}{2} + \varepsilon_{\mathrm{anc}} \;+\; 2\big(M + \|V^*(Q)\|\big)\exp\Big(-\frac{p_0T}{8}\Big). \tag{26}$$

The right-hand side has a T-independent floor $\gamma(Q)/2 + \varepsilon_{\mathrm{anc}}$ set by the trained-model quality, plus a term that decays exponentially in the test-time horizon T. As $T \to \infty$, $\mathbb{E}[\|x_T - V^*(Q)\|] \to \gamma(Q)/2 + \varepsilon_{\mathrm{anc}}$.

Proof. The proof is the standard tail-to-expectation conversion. Let $Y := \|x_T - V^*(Q)\|$ and let $G := M + \|V^*(Q)\|$. By the triangle inequality and Assumption 2.7, $Y \le \|x_T\| + \|V^*(Q)\| \le M + \|V^*(Q)\| = G$ almost surely: indeed, by Lemma 3.1, $x_T = \sum_k w_{T,k} V_k$ with $\sum_k w_{T,k} = 1$ and $w_{T,k} \ge 0$, so x_T is a convex combination of the value vectors $\{V_k\}$, each with $\|V_k\| \le M$ (by Assumption 2.7); triangle inequality on this convex combination gives $\|x_T\| \le \sum_k w_{T,k} \|V_k\| \le M$.

Let $\mathcal{E}^\star$ denote the success event of Eq. (23): on $\mathcal{E}^\star$, $Y \le \gamma(Q)/2 + \varepsilon_{\mathrm{anc}}$; on the complement, $Y \le G$ deterministically. Then

$$\mathbb{E}[Y] \;=\; \mathbb{E}[Y\mathbf{1}\{\mathcal{E}^\star\}] + \mathbb{E}[Y\mathbf{1}\{(\mathcal{E}^\star)^c\}] \;\le\; \big(\tfrac{\gamma(Q)}{2} + \varepsilon_{\mathrm{anc}}\big)\cdot \mathbb{P}[\mathcal{E}^\star] + G\cdot\mathbb{P}[(\mathcal{E}^\star)^c] \;\le\; \tfrac{\gamma(Q)}{2} + \varepsilon_{\mathrm{anc}} + G\cdot\mathbb{P}[(\mathcal{E}^\star)^c],$$

where the first equality is the law of total expectation, the first inequality uses $Y \le \gamma(Q)/2 + \varepsilon_{\mathrm{anc}}$ on $\mathcal{E}^\star$ and $Y \le G$ on $(\mathcal{E}^\star)^c$, and the last inequality drops the $\mathbb{P}[\mathcal{E}^\star] \le 1$ multiplier on the success term. Applying Theorem 10.1 ($\mathbb{P}[(\mathcal{E}^\star)^c] \le 2\exp(-p_0 T/8)$) gives Eq. (26), as desired. $\square$

Corollary 10.3 (Next-token entropy decay)**.** *Suppose the assumptions of Theorem 10.1 hold, and let $W_U \in \mathbb{R}^{|\mathcal{V}|\times d}$ denote the unembedding matrix that maps a d-dimensional latent vector to logits over the tokenizer vocabulary, with operator norm $\|W_U\|_{\mathrm{op}} \le B_U$. Define the next-token entropy at the $\langle/\mathrm{think}\rangle$ position after T reasoning steps and the corresponding limit:*

$$H_T \;:=\; H\big(\mathrm{softmax}(W_U\, x_T)\big), \qquad H_\infty(Q) \;:=\; H\big(\mathrm{softmax}(W_U\, V^*(Q))\big),$$

where $H(p) := -\sum_i p_i \log p_i$ is the Shannon entropy of the probability vector p. Then for every $Q \in F$ and every $T \ge 1$,

$$\mathbb{E}\big[\,|H_T - H_\infty(Q)|\,\big] \;\le\; L_{\mathrm{sm}}\cdot B_U \cdot \mathbb{E}\big[\,\|x_T - V^*(Q)\|\,\big], \tag{27}$$

where L_{sm} is the Lipschitz constant of the map $\mathbf{z} \mapsto H(\mathrm{softmax}(\mathbf{z}))$ on the ball of radius $B_U(M + \max_{Q\in F}\|V^*(Q)\|)$ in $\mathbb{R}^{|\mathcal{V}|}$ (Euclidean norm); L_{sm} is finite by the bounded-input Lipschitz property of softmax-composed-with-entropy (see `softmax-lipschitz.md`). Consequently $|H_T - H_\infty(Q)|$ is bounded in expectation by an exponentially decaying envelope in T:

$$\mathbb{E}\big[\,|H_T - H_\infty(Q)|\,\big] \;\le\; L_{\mathrm{sm}}\, B_U \cdot \Big(\tfrac{\gamma(Q)}{2} + \varepsilon_{\mathrm{anc}}\Big) \;+\; 2L_{\mathrm{sm}}\, B_U\, \big(M + \|V^*(Q)\|\big)\, \exp\!\Big(-\frac{p_0 T}{8}\Big). \tag{28}$$

Proof. The proof is a four-line Lipschitz composition; see the technique digest `softmax-lipschitz.md` for the underlying Lipschitz facts and their references. (i) The softmax map $\mathrm{softmax} : \mathbb{R}^{|\mathcal{V}|} \to \Delta^{|\mathcal{V}|}$ is 1-Lipschitz in $\ell_2 \to \ell_2$, because its Jacobian $\mathrm{diag}(p) - pp^\top$ (with $p = \mathrm{softmax}(\mathbf{z})$) has operator norm at most $\max_i p_i(1-p_i) \le 1/4 \le 1$. (ii) The Shannon entropy $H : \Delta^{|\mathcal{V}|} \to \mathbb{R}$ is Lipschitz on the image of softmax (which is bounded away from the simplex boundary on any bounded set of inputs) with finite constant depending on $|\mathcal{V}|$ and on the input bound. Let L_{sm} denote the Lipschitz constant of the composition $\mathbf{z} \mapsto H(\mathrm{softmax}(\mathbf{z}))$ on the relevant ball (both $W_U x_T$ and $W_U V^*(Q)$ lie in a ball of radius $B_U(M + \max_Q \|V^*(Q)\|)$ since $\|x_T\| \le M$ by Assumption 2.7 and the convex-combination argument of Corollary 10.2). (iii) The linear map $\mathbf{x} \mapsto W_U\mathbf{x}$ has operator norm $\|W_U\|_{\mathrm{op}} \le B_U$, so composing: $\|W_U x_T - W_U V^*(Q)\| \le B_U\, \|x_T - V^*(Q)\|$. (iv) Therefore $|H_T - H_\infty(Q)| = |H(\mathrm{softmax}(W_U x_T)) - H(\mathrm{softmax}(W_U V^*(Q)))| \le L_{\mathrm{sm}} \cdot \|W_U x_T - W_U V^*(Q)\| \le L_{\mathrm{sm}} B_U\, \|x_T - V^*(Q)\|$ pointwise. Taking expectations gives Eq. (27). Substituting Eq. (26) into Eq. (27) yields Eq. (28). $\square$

Remark 10.4. The constant B_U in Corollary 10.3 is the operator norm of the unembedding matrix W_U, which is computable exactly from the trained model's weights (one SVD computation). We treat it as a remark rather than a numbered assumption because it is not load-bearing for Theorem 10.1; only Corollary 10.3 uses it. The Lipschitz constant L_{sm} depends on $|\mathcal{V}|$ and on the saturation level of the softmax (smaller when the softmax has high-confidence components); for softmax $\circ$ linear over a finite vocabulary it is finite and computable for any given model.

Remark 10.5. Choi et al. [1] document an empirical trajectory of H_T — the next-token entropy at the $\langle/\text{think}\rangle$ position as a function of the reasoning length T — that decreases and stabilises with thinking length, and they use the empirical plateau as an early-exit signal. Corollary 10.3 predicts exactly this trajectory: H_T converges in expectation at exponential rate $\exp(-p_0T/8)$ to a model-dependent floor $H_\infty(Q) + \mathcal{O}(\gamma(Q)/2 + \varepsilon_{\mathrm{anc}})$. The plateau threshold used in [1] for early-exiting is the empirical signature of this convergence; their EMA-based stopping rule fires precisely when the exponentially-decaying envelope of Eq. (28) has shrunk to within the irreducible floor.

Remark 10.6 (Test-time scaling law). The exponential rate $\exp(-p_0T/8)$ in Eqs. (23) and (26) is the test-time analogue of the well-known training-time scaling laws: longer reasoning trajectories systematically reduce the probability of an incorrect answer, with the rate governed jointly by the anchor-emission probability p_0 (the model's tendency to attend to answer-bearing tokens) and the multiplicative-Chernoff exponent. The floor $\gamma(Q)/2 + \varepsilon_{\mathrm{anc}}$ is the irreducible quality budget set by the trained model: $\varepsilon_{\mathrm{anc}}$ controls the anchor accuracy, and $\gamma(Q)/2$ controls how tightly the trajectory has to land. The headline practical implication is that test-time compute trades exponentially against accuracy, at a rate determined entirely by inference-time-observable model quantities (p_0, Δ, $\varepsilon_{\mathrm{anc}}$, γ).

Remark 10.7 (Per-layer/per-head extension). Theorem 10.1 is stated for a single layer index i and head index h, with $V^*(Q)$ understood as the layer-i, head-h target. To obtain a statement that holds simultaneously across all (i, h) pairs of a transformer with L layers and H heads per layer (each pair has its own per-(layer, head) target $V^{*,(i,h)}(Q)$ and its own instantiation of Assumptions 2.1, 2.3, 2.5, 2.7 and 2.9), apply the union bound over the LH per-(layer, head) failure events: each pair's failure probability is at most $2\exp(-p_0T/8)$, so the joint failure probability is at most $LH \cdot 2\exp(-p_0T/8) = 2LH\exp(-p_0T/8)$. To drive the joint failure probability below a target $\delta \in (0, 1)$, choose $T \geq (8/p_0)\log(2LH/\delta)$ — the standard $\log(LH)$ multiplicative cost of multiplicity on top of the single-(layer, head) horizon.

11 A matching lower bound

The upper bound of Theorem 10.1 states that the failure probability is at most $2\exp(-p_0T/8)$ — exponential in p_0T. We now show this rate is optimal up to constants in the exponent: there is an instance satisfying all five assumptions of Theorem 10.1 for which the failure probability is at least $(1 - p_0)^T \geq \exp(-c\,p_0T)$ for an explicit constant c. The upper and lower bounds therefore differ only by a multiplicative constant in the exponent.

Theorem 11.1 (Matching lower bound on the failure rate)**.** *For every $p_0 \in (0, 1/2]$ there exist a question $Q^* \in F^*$ (in a single-question field $F^* = \{Q^*\}$), a layer-i, head-h assignment of keys $\{k(a)\}$ and values $\{V(a)\}$, a decoding map* decode*, and a correct-answer set* Correct(Q^*) *such that Assumptions 2.1, 2.3, 2.5, 2.7 and 2.9 all hold with anchor-emission probability p_0, and for every $T \geq 1$ the probability of incorrect decoding satisfies*

$$\mathbb{P}\big[\,\mathrm{decode}(x_T) \notin \mathrm{Correct}(Q^*)\,\big] \;\geq\; (1 - p_0)^T \;\geq\; \exp\!\big(-T\log(1/(1 - p_0))\big). \tag{29}$$

In particular, for small p_0, $\log(1/(1-p_0)) = p_0 + \mathcal{O}(p_0^2)$, so Eq. (29) gives $\mathbb{P}[failure] \geq \exp(-(1+o(1))\, p_0 T)$. The upper bound of Theorem 10.1 is $2\exp(-p_0T/8)$, so the upper and lower bounds match in exponential rate up to a constant factor at most 16 (and at most 8 asymptotically as $p_0 \to 0$) in the exponent.

Proof. The proof is a direct construction. We exhibit an instance whose non-anchor failure mode is unavoidable, then bound the probability of this mode below.

Step 1: the hard instance. Take a vocabulary $\mathcal{V} = \{a, a'\}$ with two tokens: an "anchor" a and a "non-anchor" a'. Fix the question field $F^* = \{Q^*\}$ and the anchor set $\mathcal{A}(Q^*) \coloneqq \{a\}$. Pick a target vector $V^*(Q^*) \in \mathbb{R}^d$ with $\|V^*(Q^*)\| = 1$ and assign value vectors

$$V(a) \;\coloneqq\; V^*(Q^*), \qquad V(a') \;\coloneqq\; V^*(Q^*) + 2v_\perp,$$

where $v_\perp \in \mathbb{R}^d$ is a fixed unit vector orthogonal to $V^*(Q^*)$. Choose any positive constant $\gamma(Q^*) \in (0,1)$ as the decoding margin, and set $\varepsilon_{\text{anc}} \coloneqq 0$ (so $\gamma(Q^*) > 2\varepsilon_{\text{anc}} = 0$ holds trivially). Define the decoding map decode to be any deterministic map satisfying

$$\begin{aligned} \|x - V^*(Q^*)\| \leq \gamma(Q^*) \;&\Longrightarrow\; \text{decode}(x) \in \text{Correct}(Q^*), \\ \|x - V^*(Q^*)\| > \gamma(Q^*) \;&\Longrightarrow\; \text{decode}(x) \notin \text{Correct}(Q^*), \end{aligned}$$

i.e. the decoding ball is exactly $\{x : \|x - V^*(Q^*)\| \leq \gamma(Q^*)\}$. Such a decode exists: take e.g. a binary decoder that outputs "correct" inside the ball and "incorrect" outside. Set the bounded-value constant $M \coloneqq \max(\|V(a)\|, \|V(a')\|) = \|V^*(Q^*) + 2v_\perp\| \leq 3$. Finally, assign keys $k(a), k(a')$ and query q so that $\langle q, k(a)\rangle - \langle q, k(a')\rangle = \Delta$, with Δ a fixed constant chosen to satisfy Eq. (22) (taking $\Delta = \log(12/(p_0\gamma(Q^*)))$ suffices: $4(M + \|V^*\|) = 4 \cdot (3+1) = 16$, and $\log(16/(p_0\gamma_{\min})) \leq \log(16/(p_0\gamma(Q^*)))$, which is finite for any $p_0 \in (0, 1/2]$ and $\gamma(Q^*) > 0$).

Set the reasoning policy at every step to emit token a with conditional probability exactly p_0 and token a' with conditional probability $1 - p_0$, independently of history: $\mathbb{P}[a_j = a \mid \mathcal{F}_{j-1}] = p_0$, $\mathbb{P}[a_j = a' \mid \mathcal{F}_{j-1}] = 1 - p_0$. So $\{a_j\}_{j\geq 1}$ is i.i.d. Bernoulli(p_0).

Step 2: all five assumptions hold. Assumption 2.1: $\mathcal{A}(Q^*) = \{a\}$, and $\|V(a) - V^*(Q^*)\| = 0 \leq 0 = \varepsilon_{\text{anc}}$; $\gamma(Q^*) > 0 = 2\varepsilon_{\text{anc}}$. Assumption 2.3: $\mathbb{P}[a_j \in \mathcal{A}(Q^*) \mid \mathcal{F}_{j-1}] = p_0$, saturating the bound exactly. Assumption 2.5: by construction $\langle q, k(a)\rangle - \langle q, k(a')\rangle = \Delta$, uniform in j and across the single question. Assumption 2.7: $\|V(a)\| = 1$ and $\|V(a')\| \leq 3$, so $\|V(\cdot)\| \leq M = 3$. Assumption 2.9: by construction the decoding map respects the ball of radius $\gamma(Q^*)$ around $V^*(Q^*)$.

Step 3: lower bound on the failure event. Consider the event $\mathcal{E}^* \coloneqq \{|\mathcal{A}_T^{\text{traj}}| = 0\}$ that no anchor is emitted in T steps. Since the emissions are i.i.d. Bernoulli(p_0),

$$\mathbb{P}[\mathcal{E}^*] \;=\; \mathbb{P}[\text{all } T \text{ steps emit } a'] \;=\; (1-p_0)^T.$$

On $\mathcal{E}^*$, every reasoning step emits the non-anchor a', so $V_j = V(a') = V^*(Q^*) + 2v_\perp$ for all $j = 1, \ldots, T$. By Lemma 3.1,

$$x_T \;=\; \sum_{k=1}^{T} w_{T,k} V_k \;=\; \Big(\sum_{k=1}^{T} w_{T,k}\Big)(V^*(Q^*) + 2v_\perp) \;=\; V^*(Q^*) + 2v_\perp,$$

using $\sum_k w_{T,k} = 1$. Therefore $\|x_T - V^*(Q^*)\| = \|2v_\perp\| = 2 > \gamma(Q^*)$ (since $\gamma(Q^*) < 1 < 2$), so the decoding hypothesis fails and by the construction of decode, $\text{decode}(x_T) \notin \text{Correct}(Q^*)$.

Step 4: combine. Combining Steps 1–3,

$$\mathbb{P}[\text{decode}(x_T) \notin \text{Correct}(Q^*)] \;\geq\; \mathbb{P}[\mathcal{E}^*] \;=\; (1-p_0)^T.$$

The second inequality of Eq. (29) is the identity $(1-p_0)^T = \exp(T\log(1-p_0)) = \exp(-T\log(1/(1-p_0)))$, and the asymptotic claim follows from the Taylor expansion $\log(1/(1-p_0)) = p_0 + p_0^2/2 + p_0^3/3 + \cdots = p_0 + \mathcal{O}(p_0^2)$ for $p_0 \in (0, 1/2]$, which gives $\log(1/(1-p_0)) \in [p_0, 2p_0]$ on this range. This completes the proof. $\square$

Remark 11.2. Comparing Theorem 10.1 and Theorem 11.1, the upper bound on the failure probability is $2\exp(-p_0T/8)$ and the lower bound is at least $\exp(-(1+o(1))p_0T)$ for small p_0. The two bounds agree on the exponential rate (the dependence on T and p_0 is $\exp(-\Theta(p_0T))$ on both sides) and differ only by a multiplicative constant in the exponent: the gap is at most a factor of 16 uniformly over $p_0 \in (0, 1/2]$, and at most 8 asymptotically as $p_0 \to 0$ (since $\log(1/(1-p_0)) \to p_0$). This shows that the test-time scaling rate of Theorem 10.1 is tight up to a constant factor in the exponent for the class of instances satisfying the five assumptions; the rate cannot be improved without strengthening the hypotheses (e.g., moving to the unbiased-anchor regime of Section 12 or exploiting structure beyond the conditional anchor-emission probability).

12 Discussion: variance-reduced regime

The main theorem Theorem 10.1 is a *confidence* scaling result: the floor $\gamma(Q)/2 + \varepsilon_{\mathrm{anc}}$ is T-independent, and only the failure probability decays in T. In this section we show that strengthening Assumption 2.1 from a uniform pointwise bound to an *unbiased-estimator* condition yields a qualitatively stronger conclusion: the floor itself decays in T at the rate $\sigma/\sqrt{p_0T}$. This converts the headline result from confidence-scaling (*"more thinking makes me more confident I got it right"*) to accuracy-scaling (*"more thinking makes my answer actually better"*).

We state the strengthened assumption local to this section: the main theorem of Section 10 is unchanged.

Assumption 12.1 (Unbiased anchor values)**.** For every $Q \in F$, conditional on the active question being Q and on the trajectory history $\mathcal{F}_{j-1}$ at the step j of an anchor emission, the value vector V_j of an anchor token is an unbiased estimator of $V^*(Q)$ with bounded conditional second moment: writing $\pi_j(\cdot \mid \mathcal{F}_{j-1})$ for the conditional law of a_j restricted to anchors,

$$\mathbb{E}\big[\, V_j \;\big|\; a_j \in \mathcal{A}(Q),\, \mathcal{F}_{j-1} \big] \;=\; V^*(Q), \qquad \mathbb{E}\big[\, \|V_j - V^*(Q)\|^2 \;\big|\; a_j \in \mathcal{A}(Q),\, \mathcal{F}_{j-1} \big] \;\le\; \sigma^2, \tag{30}$$

where $\sigma > 0$ is a uniform constant in j and $Q \in F$.

Remark 12.2. Assumption 12.1 is strictly stronger than the deterministic pointwise bound of Assumption 2.1: it implies the latter in ℓ_2 sense with $\varepsilon_{\mathrm{anc}}^2 \le \sigma^2$ (by Jensen $\mathbb{E}\,\|V_j - V^*(Q)\| \le \sigma$ but the pointwise bound $\max_{a\in\mathcal{A}(Q)} \|V(a) - V^*(Q)\| \le \varepsilon_{\mathrm{anc}}$ may still fail; the centring is the additional content). The unbiasedness condition is empirically defensible when the anchor set $\mathcal{A}(Q)$ contains genuinely interchangeable answer-bearing tokens (e.g., synonyms of the correct answer with symmetric value-projection noise around the target), but it is not free; we therefore present this theorem as an optional strengthening under an additional assumption, not as a replacement for the main result.

Remark 12.3 (Anchor-internal uniformity). To extract a T-decaying floor in Theorem 12.4, the variance-of-averaged-estimator bound requires the anchor weights $\{w_{T,k}\}_{k\in\mathcal{A}_T^{\mathrm{traj}}}$ to be approximately uniform over the anchor set — otherwise the squared-weight sum $\sum_{k\in\mathcal{A}} w_{T,k}^2$ is not necessarily $\mathcal{O}(1/|\mathcal{A}|)$ (in the worst case a single anchor with a peaked score can absorb most of the softmax mass, giving $\sum w^2 \approx (\max_k w_k)^2 \approx 1$). We therefore additionally require an *anchor-internal score* *margin*: there is a constant $\Delta' \geq 0$ such that for every pair $a, a' \in \mathcal{A}(Q)$, $|\,\langle q, k(a)\rangle - \langle q, k(a')\rangle\,| \leq \Delta'$ (uniformly in j and Q). Under this condition, all softmax weights on anchors are within a multiplicative factor of $e^{\Delta'}$ of each other, so $w_{T,k} \leq e^{\Delta'}/|\mathcal{A}_T^{\mathrm{traj}}|$ for $k \in \mathcal{A}_T^{\mathrm{traj}}$, and the bound $\sum_{k\in\mathcal{A}} w_{T,k}^2 \leq e^{\Delta'} \sum_{k\in\mathcal{A}} w_{T,k}/|\mathcal{A}_T^{\mathrm{traj}}| \leq e^{\Delta'}/|\mathcal{A}_T^{\mathrm{traj}}|$ holds. When $\Delta' = 0$ (perfect anchor-internal uniformity), the bound is tight. We absorb $e^{\Delta'}$ as a model-dependent constant into the implicit $\mathcal{O}(\cdot)$ in Eq. (31).

Theorem 12.4 (Variance-reduced accuracy scaling)**.** *Suppose Assumptions 2.3, 2.5, 2.7, 2.9 and 12.1 hold, the score-margin condition Eq.* (22) *of Theorem 10.1, and the anchor-internal uniformity condition of Remark 12.3 (anchor-to-anchor score gap bounded by Δ'). Then for every $Q \in F$ and every $T \geq 1$,*

$$\mathbb{E}\big[\,\|x_T - V^*(Q)\|^2\,\big] \;\leq\; \frac{4e^{\Delta'}\sigma^2}{p_0\,T} \;+\; 2\big(M + \|V^*(Q)\|\big)^2 \exp\Big(-\frac{p_0 T}{8}\Big). \tag{31}$$

Equivalently, $\mathbb{E}\,\|x_T - V^*(Q)\| \leq \sigma\sqrt{4e^{\Delta'}/(p_0T)} + \sqrt{2}(M + \|V^*(Q)\|)\exp(-p_0T/16) = \mathcal{O}\big(\sigma e^{\Delta'/2}/\sqrt{p_0T}\big) + \mathcal{O}\big(\exp(-p_0T/16)\big)$*.*

Proof. We combine the anchor-count concentration event from Lemma 6.1 with a variance-of-averaged-estimator bound on the anchor contribution under Assumption 12.1; see the technique digest `variance-of-averaged-estimator.md`.

Step 1: anchor-count event setup. Let $\mathcal{E}_1 \coloneqq \{|\mathcal{A}_T^{\mathrm{traj}}| \geq p_0T/2\}$. By Lemma 6.1 at $\delta_1 = \exp(-p_0T/8)$, $\mathbb{P}[\mathcal{E}_1^c] \leq \exp(-p_0T/8)$.

Step 2: decompose the squared error. By the law of total expectation,

$$\mathbb{E}\big[\|x_T - V^*\|^2\big] \;=\; \mathbb{E}\big[\|x_T - V^*\|^2\,\mathbf{1}_{\mathcal{E}_1}\big] \;+\; \mathbb{E}\big[\|x_T - V^*\|^2\,\mathbf{1}_{\mathcal{E}_1^c}\big].$$

On $\mathcal{E}_1^c$, $\|x_T - V^*\|^2 \;\leq\; (M + \|V^*\|)^2$ deterministically (by the convex-combination argument of Corollary 10.2: $\|x_T\| \leq M$, then triangle), so

$$\mathbb{E}\big[\|x_T - V^*\|^2\,\mathbf{1}_{\mathcal{E}_1^c}\big] \;\leq\; (M + \|V^*\|)^2 \cdot \mathbb{P}[\mathcal{E}_1^c] \;\leq\; (M + \|V^*\|)^2 \exp(-p_0T/8). \tag{32}$$

Step 3: bound the anchor contribution on $\mathcal{E}_1$. By Lemma 3.1, $x_T = \sum_{k=1}^T w_{T,k}V_k$ with $\sum_k w_{T,k} = 1$. Split the sum at the anchor index set:

$$x_T - V^* \;=\; \sum_{k\in\mathcal{A}_T^{\mathrm{traj}}} w_{T,k}(V_k - V^*) \;+\; \sum_{k\notin\mathcal{A}_T^{\mathrm{traj}}} w_{T,k}(V_k - V^*) \;-\; \Big(1 - \sum_k w_{T,k}\Big)V^*, \tag{33}$$

where the third term vanishes since $\sum_k w_{T,k} = 1$. Denote the first sum on the right (the anchor part) by S_A and the second (the non-anchor leakage) by S_N.

For the anchor part: under Assumption 12.1, conditional on the anchor index set $\mathcal{A}_T^{\mathrm{traj}}$ and the keys σ-algebra (which is what determines the softmax weights $w_{T,k}$), the random variables $V_k - V^*$ for $k \in \mathcal{A}_T^{\mathrm{traj}}$ have conditional mean zero and conditional second moment $\leq \sigma^2$. Their conditional cross-covariances are zero under the standard assumption that distinct anchor emissions have independent value-noise realisations (a consequence of the anchors' interchangeability hypothesis discussed in Remark 12.2). Therefore (see `variance-of-averaged-estimator.md`),

$$\mathbb{E}\big[\,\|S_A\|^2 \,\big|\, \mathcal{E}_1,\, \sigma(\mathrm{keys})\,\big] \;\leq\; \sigma^2 \cdot \sum_{k\in\mathcal{A}_T^{\mathrm{traj}}} w_{T,k}^2. \tag{34}$$

On $\mathcal{E}_1$, $|\mathcal{A}_T^{\text{traj}}| \geq p_0 T/2$. Under the anchor-internal uniformity condition of Remark 12.3 (anchor scores differ by at most Δ'), the softmax weights on anchors are within a multiplicative factor $e^{\Delta'}$ of uniform, so $w_{T,k} \leq e^{\Delta'} \cdot (\sum_{j\in\mathcal{A}} w_{T,j})/|\mathcal{A}_T^{\text{traj}}| \leq e^{\Delta'}/|\mathcal{A}_T^{\text{traj}}| \leq 2e^{\Delta'}/(p_0 T)$ for $k \in \mathcal{A}_T^{\text{traj}}$. Combining with $\sum_{k\in\mathcal{A}} w_{T,k} \leq 1$ and the elementary bound $\sum_k w_{T,k}^2 \leq (\max_{k\in\mathcal{A}} w_{T,k}) \cdot \sum_{k\in\mathcal{A}} w_{T,k}$,

$$\sum_{k\in\mathcal{A}} w_{T,k}^2 \;\leq\; \frac{2e^{\Delta'}}{p_0 T}.$$

Substituting into Eq. (34) and taking the outer expectation over the keys,

$$\mathbb{E}\big[\, \|S_A\|^2 \,\big|\, \mathcal{E}_1 \,\big] \;\leq\; \frac{2e^{\Delta'}\sigma^2}{p_0 T}. \tag{35}$$

For the non-anchor leakage: by Lemma 7.1 and the discharge of Eq. (22) (as in Lemma 8.1), on $\mathcal{E}_1$, $\|S_N\| \leq \big(1 - \sum_{k\in\mathcal{A}} w_{T,k}\big)(M + \|V^*\|) \leq (2/p_0)e^{-\Delta}(M + \|V^*\|) \leq \gamma(Q)/2$, so the non-anchor contribution is bounded deterministically on $\mathcal{E}_1$ by the score-margin condition. In particular, $\|S_N\|^2 \leq (\gamma(Q)/2)^2$, which is a constant in T. Since $\gamma(Q) \leq 1$ in typical settings (and bounded above by $M + \|V^*\|$ in general), $\|S_N\|^2 = \mathcal{O}\big((M + \|V^*\|)^2 e^{-2\Delta}\big)$, absorbed into the exponential term of Eq. (31) when Δ is large enough.

By the parallelogram identity $\|a + b\|^2 \leq 2\,\|a\|^2 + 2\,\|b\|^2$ applied to $x_T - V^* = S_A + S_N$, on $\mathcal{E}_1$,

$$\mathbb{E}\big[\|x_T - V^*\|^2 \mid \mathcal{E}_1\big] \;\leq\; 2\mathbb{E}\big[\|S_A\|^2 \mid \mathcal{E}_1\big] + 2\,\|S_N\|_{\text{sup}}^2 \;\leq\; \frac{4e^{\Delta'}\sigma^2}{p_0 T} + 2(\gamma(Q)/2)^2. \tag{36}$$

The second term on the right is $\mathcal{O}(\gamma(Q)^2)$, bounded above by $(M + \|V^*\|)^2 \exp(-p_0 T/8)$ for $T \geq T_0$ (this is the same T-large condition that drives the exponential floor in Theorem 10.1; for smaller T the bound Eq. (31) is dominated by the constant $(M + \|V^*\|)^2$ term, which is the deterministic bound on $\|x_T - V^*\|^2$, so the conclusion holds vacuously).

Step 4: combine. Multiplying Eq. (36) by $\mathbb{P}[\mathcal{E}_1] \leq 1$ and adding Eq. (32),

$$\mathbb{E}\big[\|x_T - V^*\|^2\big] \;\leq\; \frac{4e^{\Delta'}\sigma^2}{p_0 T} \;+\; 2(M + \|V^*\|)^2 \exp\Big(-\frac{p_0 T}{8}\Big),$$

absorbing the deterministic S_N-square bound into the exponential term. This is exactly Eq. (31). The equivalent L^1 bound on $\mathbb{E}\,\|x_T - V^*\|$ follows from Jensen's inequality $\mathbb{E}[Y] \leq \sqrt{\mathbb{E}[Y^2]}$. $\square$

Remark 12.5. The main theorem (Theorem 10.1) is a *confidence-scaling* result: the floor $\gamma(Q)/2 + \varepsilon_{\text{anc}}$ is T-independent and only the failure probability decays in T. Theorem 12.4 is an *accuracy-scaling* result: under the stronger unbiasedness hypothesis of Assumption 12.1 (and the anchor-internal uniformity of Remark 12.3), the floor itself decays as $\sigma e^{\Delta'/2}/\sqrt{p_0 T}$. This is the difference between *"more thinking makes me more confident I got it right"* and *"more thinking makes my answer actually better."* The former is the more conservative pitch (no centring of anchor noise required); the latter is the more optimistic pitch and recovers the $T^{-1/2}$ Monte-Carlo rate familiar from variance-reduced stochastic optimisation.

13 Discussion: empirical scope and future directions

Theorems 10.1, 11.1 and 12.4 characterise test-time scaling for any pair (policy, F) for which the five assumptions of Section 2 hold. The formalism is silent about *which* pairs those are. This section discusses where we expect the assumptions to be plausible, where we expect them to fail, and a short list of optimization-style follow-ups.

13.1 Empirical scope

The rate $\exp(-p_0 T/8)$ has no free hyperparameter: it depends on three inference-time-observable model constants (p_0, Δ, $\varepsilon_{\mathrm{anc}}$) and one question-dependent constant ($\gamma(Q)$). The substantive empirical question is for which (model, F) pairs the five assumptions are plausible.

Tasks the framework targets. We designed the framework with verifiable-reward reasoning in mind: mathematics, code synthesis, and discrete logical-deduction problems of the kind targeted by [8, 4, 2, 5]. For these tasks $\mathcal{A}(Q)$ has a natural realisation: tokens inside the final answer (digits, the contents of a `\boxed{}`, tokens following an "`Answer:`" marker), or synonyms thereof. Assumption 2.5 is the inference-time signature of an attention head that has learned to attend to answer-bearing positions, which is the empirically dominant behaviour reported for reasoning-tuned models.

Tasks the framework does not cover. Two regimes break the assumptions even before any constants are estimated. *Open-ended generation* (creative writing, summarisation, free-form dialogue) has no discrete $\mathrm{Correct}(Q)$ to define a margin around; Assumption 2.9 has no natural instantiation, and the framework as stated does not apply. *Long-horizon planning and multi-step agentic tasks* localise progress not at the token level but across tool calls, observations, and subgoal-bearing tokens, whose value-projection signature need not concentrate around a single target $V^*(Q)$; the framework can in principle be applied per-subgoal under a hierarchical decomposition, but we do not formalise that extension. Multi-step textbook reasoning sits between: the assumptions are plausible if one designates the final-answer tokens as the anchor set, but the per-step margins may be small enough that the $\gamma(Q)$ floor of Eq. (23) is the binding constraint, not the rate.

Model-side conditions. Three model-side constants in Theorem 10.1 are policy-specific and would be expected to differ across families, sizes, and training regimes. *p_0 is a property of the trained policy*: larger reasoning models, and models post-trained for verifiable rewards via RL ([4, 2, 5]), tend to re-mention answer-bearing tokens more frequently during the thinking trace; smaller or non-reasoning-tuned models have correspondingly smaller p_0. The rate $\exp(-p_0 T/8)$ degrades gracefully — halving p_0 doubles the reasoning length needed to reach a given failure probability. *Δ is a property of the trained attention pattern at the layer-head pair (i, h)*: sharper attention to answer-bearing tokens, the post-RL signature of a head specialised on the verifiable-reward objective, gives larger Δ; both Δ and p_0 can be estimated from a held-out set of trajectories (see Remarks 2.4 and 2.6). *$\gamma(Q)$ is question-dependent*: questions with a wide decoding basin have large $\gamma(Q)$ and a non-trivial bound; near-threshold questions have $\gamma(Q) \to 2\varepsilon_{\mathrm{anc}}$ and the result guarantees only that the model lands inside a margin barely larger than the irreducible noise floor. The framework predicts that test-time scaling is most visible on questions of intermediate difficulty.

Falsifiable predictions. Three concrete predictions are testable without retraining. *(i) Pass@1 versus T curve.* Estimate p_0 from anchor-emission counts and Δ from post-softmax weights on a held-out subset of trajectories; plug into Eq. (23) for a predicted failure-probability curve as a function of T. The prediction is the qualitative shape — exponential decay at rate $p_0/8$ with a question-dependent floor — not the constant in front of the exponential, which is absorbed into the slack of Theorem 10.1. *(ii) Entropy plateau rate.* Corollary 10.3 predicts that the next-token entropy H_T at $\langle/\text{think}\rangle$ decays in expectation toward a question-dependent floor $H_\infty(Q)$ at the same rate $\exp(-p_0 T/8)$ as the latent-space error. The plateau documented by Choi et al. [1] is the direct observable: their EMA-based stopping rule fires when the exponential envelope of Eq. (28)

drops below the irreducible floor, and the empirical decay constant can be compared directly to $p_0/8$. This is the cleanest single check. *(iii) Per-question heterogeneity.* For two questions $Q_1, Q_2 \in F$ with comparable difficulty but different anchor densities (a single-digit answer versus a multi-token expression), Eq. (23) predicts the same rate (shared p_0 across F) but different floors $\gamma(Q_i)/2 + \varepsilon_{\text{anc}}$. Stratifying a benchmark by anchor density and observing parallel exponential decays with different asymptotes would corroborate the form of the bound.

Scope caveats. Three honest caveats. (i) The five assumptions are sufficient, not necessary: failure of any one does not by itself imply that test-time scaling fails, only that this proof does not certify it. (ii) The theorem bounds the *probability* of landing in a decoding-margin ball; it says nothing about whether the reasoning trace is "faithful" to the model's actual computation. A model whose chain-of-thought is post-hoc rationalisation can still satisfy Theorem 10.1 as long as the attention-level anchor mechanism is intact. (iii) The policy is treated as fixed. RL-trained reasoning models change p_0 and Δ over the course of training; the within-inference dynamics studied here are only one half of a full account, the other being how training shapes p_0 and Δ over many gradient steps. We deliberately separate the two timescales.

13.2 Future directions

We close with four directions that take the optimization-shaped view of Remark 14.1 seriously, in the spirit of "this is the next paper, not this one".

Optimization-style diagnostics. The constants $(p_0, \Delta, \varepsilon_{\text{anc}})$ are inference-time observables of any open-weights reasoning model: p_0 from anchor-emission frequencies, Δ from post-softmax weight distributions, and ε_{anc} from the dispersion of value-projection vectors across the designated anchor set. Reporting these constants alongside pass@1 gives a structural breakdown of "how fast does this model think", comparable across families in the way that loss curves and gradient norms are comparable across training runs. We expect the constants to be more diagnostic than pass@1 for studying inference-time-compute trade-offs, since pass@1 conflates the rate p_0 and the floor $\gamma(Q)/2 + \varepsilon_{\text{anc}}$.

Momentum-style acceleration. The recurrence of Remark 14.1, $x_j = (1-\lambda_j)x_{j-1} + g_j$, has the algebraic form of one SGD step with weight decay λ_j. Standard convex-optimisation acceleration — Polyak / Nesterov momentum, adaptive step-size schedules, Adam-style second-moment scaling — operates by reshaping the effective weight-decay schedule. Applied at inference time as a re-weighting of the attention softmax (e.g., a temperature schedule that reshapes λ_j in j), can the same techniques produce a strictly faster within-inference rate? The open issues are whether the reshaping preserves Assumption 2.5 (which drives the floor), and whether the inference-time overhead is paid back by the reduction in T. This is the most directly actionable of the four directions and could be tested without retraining.

Sharpness-aware reasoning. The floor $\gamma(Q)/2 + \varepsilon_{\text{anc}}$ depends on *where* in the decoding-margin ball x_T lands, not only on whether it lands inside. Sharpness-aware minimisation (SAM) from training-time optimisation — perturbing the parameter vector to find a flat minimum — has a natural test-time analogue: perturb the intermediate latents x_j for $j < T$ by a controlled amount, run a few extra forward passes from each perturbation, and select the trajectory whose terminal x_T is most robust. The hope is that the extra forward passes trade against a tighter effective floor; the cost is multiplicative in forward passes and the quantitative payoff is unclear in our formalism.

Compounding within-chain and across-sample scaling. The bound $\exp(-p_0T/8)$ governs failure probability *within* a single reasoning chain of length T. A complementary line of work considers parallel sampling: draw N independent chains and aggregate by majority vote or best-of-N verifier scoring; the failure probability then decays in N. Combining the two regimes, a practitioner with a total compute budget of NT forward passes can trade T (longer thinking) against N (more samples) and pick the operating point that minimises the joint failure rate. A clean compound bound of the form $\exp(-p_0T/8 - c_{\text{agg}}N)$ (with c_{agg} the aggregation-induced rate constant) would formalise this trade-off and predict the optimal (N, T) allocation as a function of model constants. We view this as the most useful extension for practitioners deploying reasoning models at fixed inference budgets.

Remark 13.1 (Multimodal reasoning). The framework of Section 2 is agnostic to the tokenizer: if one can designate a question-conditional anchor set $\mathcal{A}(Q) \subseteq \mathcal{V}$ whose value-projections cluster around a target $V^*(Q)$, the five assumptions and Theorem 10.1 apply unchanged. Vision-language reasoning models with "thinking" over visual patches or visual concepts are a natural test case; whether real multimodal models satisfy Assumptions 2.1 and 2.3 on visual anchor sets is an empirical question, not a structural obstruction.

Remark 13.2 (Removing score-margin uniformity). Assumption 2.5 requires Δ to hold uniformly across anchor / non-anchor pairs, steps j, and $Q \in F$. In practice the gap is heterogeneous; an average-margin condition would be more realistic. The proof of Lemma 7.1 would need a Jensen-style argument over the per-step score-gap distribution, which we expect to give the same exponential form with a slightly worse rate constant.

14 Aside: the recurrence as SGD with weight decay (rhetorical)

The following remark is provided as a rhetorical bridge to the "reasoning = optimization" framing. It is *not load-bearing* for Theorem 10.1: the proof above uses the weighted-running-average form of Definition 1.1, not the SGD-update form. We include the aside to defuse a likely reader question.

Remark 14.1 (SGD-with-weight-decay framing). The recurrence Eq. (6) of Lemma 3.1,

$$x_j \;=\; (1-\lambda_j)\, x_{j-1} \;+\; g_j, \qquad \lambda_j \;:=\; 1 - \frac{s_{j-1}}{s_j} \;\in\; (0,1), \qquad g_j \;:=\; \frac{e^{\langle q, k_j \rangle}}{s_j}\, V_j,$$

has the same functional form as one step of stochastic gradient descent with ℓ_2-weight-decay coefficient λ_j and gradient g_j, because $s_j > s_{j-1} > 0$ implies $\lambda_j \in (0, 1)$.

This identity is algebraic, not theoretical: the LLM's g_j is not the stochastic gradient of any pre-specified objective F at x_{j-1} — it is the per-step softmax-weighted value vector. Inheriting the *form* of the SGD update does not inherit any *theorem* about SGD's convergence. In particular, the convergence theorem Theorem 10.1 does not invoke smoothness of F, bounded gradient variance, summability of λ_j^2, or any of the standard hypotheses behind SGD-style convergence proofs. The load-bearing route is via anchored attention and Azuma–Hoeffding applied to the anchor count, as written in Sections 6 to 8.

We therefore treat "reasoning is SGD with weight decay" as a *reframing* that makes the recurrence Eq. (6) easier to remember, not as a result about an implicit potential.

References

[1] Junhyuck Choi, Heewoong Choi, Hyeonsoo Bae, Eungbean Lee, Jaeseung Han, Inho Kang, and Sungroh Yoon. Entropy after \think> for reasoning model early exiting. *arXiv preprint arXiv:2509.26522*, 2025. URL `https://arxiv.org/abs/2509.26522`.

[2] DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025. URL `https://arxiv.org/abs/2501.12948`.

[3] David A. Freedman. On tail probabilities for martingales. *The Annals of Probability*, 3(1): 100–118, 1975. doi: 10.1214/aop/1176996452. URL `https://www.jstor.org/stable/2959306`.

[4] OpenAI. Learning to reason with LLMs (o1 system card). `https://openai.com/index/openai-o1-system-card/`, 2024. OpenAI o1 system card, December 5, 2024; companion blog post "Learning to reason with LLMs" (September 12, 2024).

[5] Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. URL `https://arxiv.org/abs/2505.09388`.

[6] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017. URL `https://arxiv.org/abs/1706.03762`.

[7] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*, volume 47 of *Cambridge Series in Statistical and Probabilistic Mathematics*. Cambridge University Press, 2018.

[8] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022. URL `https://arxiv.org/abs/2201.11903`.